Ezequiel Boero <ezequiel.boero@adagioinstitute.org>

Re: Koopman Spectral Theory Paper (2) - My Response Your Latest Comments

Benjamin D. Summers <bsummers@adagiollc.com> Fri, Sep 25, 2026 at 3:26 PM
To: Ezequiel Boero <ezequiel.boero@adagioinstitute.org>
Cc: "ISMAIL-SUTTON, SARA R." <sara.r.ismail-sutton@durham.ac.uk>

Hi Ezequiel,

Thank you for re-deriving the new claims. Revision 6.1 of the manuscript is up at SSRN (https://doi.org/10.2139/ssrn.6733181) and makes the two amendments.

Table 1, first row: Theorem 4.8 is now listed separately, with the statement that under the cocycle the invariant sum $\mathfrak{c} + \mathfrak{d} = \Delta$ vanishes while the gauge-dependent summands need not, citing Theorem 4.8(c) and Remark 4.9. The other four results keep "each object vanishes identically under the cocycle."
Discussion, fourth open question: "approached only by families whose intermediate kernels are deterministic" is replaced by "approached by the extremal families exhibited in Proposition 4.7, whose intermediate kernels are deterministic." I have no rigidity argument, so the subclass question that follows stays open as the precise statement of what is missing.

Nothing else changed. Your correction on attainability matches the manuscript's wording, and the "leading direction v ⊗ v" sentence in my last email was loose where the manuscript is not.

Ben

Benjamin D. Summers
Managing Director
ADAGIO GROUP
MOBILE +1.713.775.1011 | **OFFICE** +1.305.433.1823 | **FAX** +1 239.932.5558
bsummers@adagiollc.com | adagio.capital

On Fri, Sep 25, 2026 at 6:38 AM Ezequiel Boero <ezequiel.boero@adagioinstitute.org> wrote:

> Ben,
>
> Thank you for the update and for the responses to the questions raised in my report.
>
> I have checked the changes against the last deposited version of the article; I re-derived and controlled every new claim rather
> than checking your responses against the reports.
> My assessment is favourable: the mathematical, interpretive and priority issues raised in the two reports have in substance been addressed.
> Two narrow textual amendments remain, set out at the end.
>
> **Proposition 4.7.** This is the substantial item, and it is right. The manuscript is stronger than your email: the continuous-time family
> $\Gamma(t \leftarrow s) = I - D\psi(t-s)vv^*$ with $\psi(\tau) = \max\{0, 2\tau/T - 1\}$ makes the statement hold on the whole simplex, and the proof of (a) replaces the heuristic that
> the product term opposes the deficit with a clean reduction: the rank-one form $A = I + vw_A^*$, so $\|I - A\| = \|w_A\|$ and $\langle w_A, v\rangle = -(p_A + q_A)$;
> the composition $AB = I + v(w_A + (1 - \sigma_A)w_B)^*$; and then $\|\Gamma(T \leftarrow 0) - AB\| \geq \langle Dv + w_A + (1 - \sigma_A)w_B, v\rangle = D - \sigma_A - (1 - \sigma_A)\sigma_B$,
> nonincreasing in both, giving $D - 2\varepsilon + \varepsilon^2$ and hence $\varepsilon^2 - 3\varepsilon + D \leq 0$.
> Every step checks, and the lower bound applies to arbitrary column-stochastic comparison matrices, not merely to the symmetric reduction.
> I confirmed independently that m(D) is the global minimum over the full four-parameter class for D from 0.2 to 1.9; that your semigroup attains
> it with the supremum over all pairs, not merely the three of the triple, with $\|\tilde{P}\| = 1$; and that the monotonicity step in (b), which needs
> $-(1 - m)\ln(1 - m) \leq D$, holds up to D = 1.99.

The identities m = D/3 + m²/3 and m/(D/3) = 1 + m²/D are exact, and the five ratios of my Finding 4 reproduce.
The proposition closes the former open remark and settles Question 2.

A correction to my own report. My Finding 4 said the constant is optimal "with equality requiring both intermediate kernels to be deterministic"
and offered the three-point family as "an extremal family." That wording implies attainability, and your proposition shows there is none:
the optimum for that family is m(D) > D/3 strictly for every D > 0, so no family attains 1/3 and the constant is optimal only as an infimum
approached as D → 0. My conclusion stands; my formulation was imprecise, and the manuscript's phrasing (cannot be improved, approached though not attained)
is the correct one. I should also record that my derivation was a first-order argument inside the two-state symmetric reduction.
The exact value, the extension to all column-stochastic comparisons, the continuous-time family and the strictness are yours.

A point I raised and now withdraw. On your email I was going to object that "leading direction v ⊗ v" reads falsely, since the leading
singular direction of the product deficit is v ⊗ v only when α(t,s) + α(t,r)α(r,s) > 1 (true near the diagonal, false on wide triples, where at (0, ½, 1)
with λ = 0.7 and ε = 0.9 the singular values are (0.2486, 0.2486, 0.0865, 0) and v ⊗ v is the weakest nonzero direction).
The manuscript does not share the looseness: Section 6.8 gives D_⊗ = |δ| max{1, α(t,s) + α(t,r)α(r,s)}, which is exactly that criterion, and attaches
the leading direction to β, where there is no ambiguity. I verified the diagonal form, the closed-form norm, the singular values 4ε, 2ε, 2ε, 0,
the rank-three and mean-zero statements, the 2^k Kronecker-sum iteration, and the agreement with the scaling law near the diagonal.
The objection was to a sentence in correspondence, not to the paper, and I withdraw it.

The completion argument. The revised proof resolves the most delicate point from Sara's review.
P(t,s) = V(t,s)V(t,s)* has constant rank on the closed simplex, including the diagonal, so the bundle is trivial over a contractible base;
rotating a continuous frame by the diagonal unitary G(s)$^{-1}$ pins it to the reference frame at t = s, and the completion is continuous with W(s,s) = I exactly.
That is the relative trivialisation the earlier argument had not supplied, and since the rotation depends on s alone it preserves joint continuity in (t,s).
It is valid in the finite-dimensional setting where it is used.

Priority, attribution and interpretation. All satisfactory. The identity C = A′ + A² is attributed to the Dyson/Magnus setting.
Canturk and Doukas are cited and Section 5 no longer claims the basic lift. Remark 5.7 states that a formal Hamiltonian generator can be constructed
for a selected continuous dilation and records the gauge law $\hat{H} \mapsto (I\otimes U)\hat{H}(I\otimes U)^* + i(I\otimes \dot{U}U^*)$; since $\dot{U}U^*$ ranges over all anti-Hermitian functions the additive term is an arbitrary Hermitian function, so your conclusion that no feature of $\hat{H}$ is invariant is not merely safe but exact, and the invariant content is correctly assigned to the channel family, the populations and the deficit.
The sewing paragraph, the three proportion labels (Proposition 3.6, Theorem 4.5(a), Corollary 4.6) and the Theorem 5.4 lead-in separating imported
machinery from the diagonal-retention indexing are all in place, and no broad priority language remains.
Sara's contraction qualification now appears at all four locations.
The augmentation passage confines the Bell/Fine and Kochen–Specker exclusions to Bell-local and noncontextual augmentations and does not infer
general non-augmentability; both attributions are correct. Table 1 answers Sara's request in the register she specified.
Your answer to Question 7 is a useful concrete protocol and correctly identifies the h^4 penalty of resolving a second-order quantity:$N \gtrsim (\rho h^2 \|\beta\|)^{-2}$.
Since the estimator and its bootstrap theory remain undeveloped, retaining it as an open methodological problem is the right boundary.

Two narrow amendments remain.

1. Table 1 overstates what vanishes under the cocycle in Theorem 4.8. The first row says the objects associated with Theorem 4.8 vanish
identically under the cocycle. Theorem 4.8(c) and Remark 4.9 say otherwise, and the counterexample is yours:
for the divisible triple Γ(t,s) = Γ(t,r) = Γ(r,s) = F, the gauge (H, R(45°), R(45°)) gives V = R(90°), hence D = F − X and C = X − F,
both of maximum modulus ½, with exact cancellation, while (KH, K, H) annihilates both separately.
I recomputed C from definition (8) rather than by difference, and it is as the remark states.
What vanishes invariantly under CK is the sum, C + D = Δ = 0, not the summands (which is also what the lead-in to

that section says, that the individual
terms are properties of the selected lifts and not of the family).
Please revise the row to say that the invariant sum vanishes under CK while the gauge-dependent summands need not.

2. The exclusivity clause following Proposition 4.7 is stronger than the proof. The fourth item of the Discussion says the constant is approached
"only by families whose intermediate kernels are deterministic." Proposition 4.7 proves that the exhibited extremal family has deterministic intermediate kernels;
it does not classify every asymptotically extremising family. There is a further reason to weaken it: the same sentence declares open "whether it can be
improved on subclasses, for instance families whose intermediate kernels are bounded away from deterministic," and those two halves are the same
question: the exclusivity clause, in any quantitative form, implies that no family in that subclass approaches 1/3, which is precisely the improvement there declared open, and conversely. As written the sentence asserts as established the qualitative form of what it then poses as a problem.
Either supply the rigidity argument or replace "only by" with something like "by the extremal families exhibited in Proposition 4.7,
whose intermediate kernels are deterministic," leaving the subclass question as the precise statement of the missing rigidity.

Neither amendment affects the construction, the proof of sharpness, or the principal conclusions.

In summary, subject to those two amendments, I regard the questions raised in both technical reports as answered, withdrawn where appropriate,
or correctly recorded as limitations and open problems, and I have no further substantive revision to ask for on this paper.

Ezequiel

On Thu, Sep 24, 2026 at 1:22 AM Benjamin D. Summers <bsummers@adagiollc.com> wrote:

> Hi Sara and Ezequiel,
>
> Below I first address Sara's comments on my response to her manuscript review, then below I address Ezequiel's review as I had failed to previously. Thank you both for your efforts.
>
> *On Sara's comments...*
>
> Thank you for the follow-up. Revision 5 (https://dx.doi.org/10.2139/ssrn.6733181) implements the four open points.
>
> The 1/3 bound now carries the contraction assumption at every statement, including the two sentences that lacked it (the lead-in to Corollary 4.6 and the Discussion summary).
>
> The augmentation passage names the constraints and the theorems: Bell locality, excluded by Bell's theorem in Fine's form (a joint distribution over all settings exists iff CHSH holds); noncontextuality, excluded by Kochen–Specker in dimension at least three. Bell, Fine, and Kochen–Specker are cited; the sentence that the paper does not establish non-augmentability stays.
>
> Theorem 4.5(a) is now marked, in the paragraph before it, as a structural consequence of the subspace definitions and the Schmidt decomposition, with the dynamical content located in (b), Corollary 4.6, Proposition 4.12, and Theorem 7.4.
>
> Table 1 in the related-work section gives the theorem-level comparison: for each of the three programs, what it presupposes, which results here have a counterpart in it, and which have none. If it misses a comparison you had in mind, tell me which.
>
> On Step 1b: by (R1), P(s,s) is exactly the projection onto $H\otimes|0\rangle$, so $I - P(t,s)$ has constant rank on the closed simplex; Steenrod triviality over the contractible base gives a continuous frame of its range; rotating that frame by $G(s)^{-1}$, where $f_\alpha(s,s) = \Sigma_\beta G_{\beta\alpha}(s)\, e_\beta$ with G continuous in s, makes it equal the reference frame on the diagonal, so $W(s,s) = I$.
>
> Thank you for withdrawing the (R3′) point.
>
> *On Ezequiel's review...*
>
> Thank you for the evaluation. Your report examined the September 3 deposit; revision 6, enclosed, post-date it.

Revision 5 (https://dx.doi.org/10.2139/ssrn.6733181) implements every item except the title, which stays, and Question 7, which I answer below rather than in the paper.

Finding 4, Question 2. Confirmed. For the family $\Gamma(T,T/2) = \Gamma(T/2,0) = I$, $\Gamma(T,0) = I - (D/2)(I - X)$, every divisible column-stochastic comparison has $\sup \delta \geq m(D) = (3 - \sqrt{(9 - 4D)})/2$, and the semigroup with $e^{-\kappa T/2} = 1 - m(D)$ attains it; $m(D)/(D/3) = 1 + m(D)^2/D$, which reproduces your five ratios and tends to 1. The lower bound holds for all column-stochastic comparisons, not only the symmetric reduction: a column-stochastic 2×2 matrix has $A - I = v\, w^{\mathsf{T}}$ with $\langle w, v\rangle \leq 0$, so the product term of AB opposes the deficit for every admissible pair. The result is Proposition 4.7, with the family extended to continuous time so that the statement holds on the whole simplex, and the former Remark 4.7 is closed. The bound is strict for every $D > 0$ and is approached only at deterministic intermediate kernels; the paper says so.

Finding 1, Question 1. The identity $C = A' + A^2$ for $\partial_t U = AU$ is stated with Dyson and Blanes–Casas–Oteo–Ros cited, followed by a sentence saying what the section adds: the residual as a named invariant, the factor $(t - r)(r - s)$, the singular-direction reading, and the place of β in the global theory.

Finding 5, Question 4. Canturk and Doukas are cited in the related-work section and at the sixth contribution, and Section 5 no longer claims the basic lift. What remains is the diagonal-retention channel with $\Phi(s,s) = \mathrm{id}$, the jointly continuous completion with $W(s,s) = I$ on a fixed environment of dimension $n^2 - n + 1$, the formal generator under (R3′), the grid cocycle, and the incorporation into the deficit framework.

Finding 2, Question 5. The remark now reads "a formal Hamiltonian generator can be constructed for a selected continuous dilation" and states the gauge dependence: under $W \mapsto (I \otimes U(t))W$ the channel is unchanged and $\hat{H}$ transforms by conjugation plus $i\dot{U}U^*$, so no feature of $\hat{H}$ is invariant; the invariants are the channel family, the populations, and the deficit.

Proportion and priority. Proposition 3.6 and Corollary 4.6 carry the scope-fixing label already used for Proposition 2.6, Theorem 5.4 has a lead-in separating the imported machinery (Stinespring, Kretschmann–Schlingemann–Werner, Kato, Steenrod) from the diagonal-retention indexing, and the sewing and almost-flow literature is acknowledged in a new related-work paragraph.
Question 3. The nearest notions I know are the Sz.-Nagy–Foiaş defect spaces of a single contraction, the sewing defect $\mu_{ts} - \mu_{tu} \circ \mu_{us}$ of almost-flows, and Tian's defect kernel of a compressed semigroup (June 2026). None spans the ranges of all composition deficits over all triples; that global locus, with vanishing on $V_{in}^{\perp}$ and the approximation obstruction of Theorem 4.5(b), is the claim I would defend. The pointwise range and kernel facts are linear algebra.

Question 6. Theorem 5.4(iii) is a collision model with a fresh environment factor per grid increment, and Table 1 lists which results have process-tensor counterparts. The distinctions are classical input, a fixed environment per pair, and joint continuity in (t,s).

Question 7. Estimate $\hat{\Gamma}$ on a time grid from paired observations, form $\hat{\Delta}$ on triples, and test $\Delta \equiv 0$ by parametric bootstrap under the fitted product family, with statistic max over triples of $\hat{D}$; estimate β from the slope of $\hat{\Delta}$ against $(t - r)(r - s)$ over near-diagonal triples rather than from second differences of $\hat{\Gamma}$. Conditioning: $\Delta \sim h^2\|\beta\|$ while $\hat{\Gamma}$ carries error of order $N^{-1/2}$, so relative precision ρ on β needs $N \gtrsim (\rho\, h^2\, \|\beta\|)^{-2}$ transitions per pair; the h^4 is intrinsic to a second-order quantity, and the regression removes only the cancellation between separately estimated coefficients. The paper keeps the estimator open.
Question 8. A measured deficit establishes non-divisibility of the chosen description and the absence of a Dyson-class generator for it, not intrinsic memory: any finite-memory process is a lumping of a Markov process on an augmented space, and a compressed semigroup has a nonzero defect. Remark 2.8 and Section 8 say so.

Question 9. New Section 6.8: the theorems carry no dimensional hypothesis, the two examples witness nonvacuity in closed form, and a product rule, $\Delta_{\Gamma\otimes\tilde{\Gamma}} = \Delta \otimes \tilde{\Gamma}(t,s) + \Gamma(t,r)\Gamma(r,s) \otimes \tilde{\Delta}$ and $\beta_{\Gamma\otimes\tilde{\Gamma}} = \beta \otimes I + I \otimes \tilde{\beta}$, supplies closed-form families in dimension 2^k. The product of two copies of the two-state family is a four-state family with a rank-three deficit, $V_{in} = V_{out}$ the mean-zero subspace, and leading direction $v \otimes v$.

Ben

Benjamin D. Summers
Managing Director
ADAGIO GROUP
MOBILE +1.713.775.1011 | **OFFICE** +1.305.433.1823 | **FAX** +1 239.932.5558
bsummers@adagiollc.com | adagio.capital